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COMPLETE NOTES

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**For Hamromaster Online
Classes**

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MEASURE OF CENTRAL TENDENCY

Short Questions [2071-2079]

2072(I) The following are the ages of the 8 people in the video arcade at the City Shopping Mall at 2:00 P.M.

10, 12, 10, 13, 14, 15, 8, 10

Determine (i) mean (ii) median (iii) mode (iv) standard deviation (v) range of ages.

Solution,

Given values are: 10, 12, 10, 13, 14, 15, 8, 10

① Mean: $\Sigma x = 10 + 12 + 10 + 13 + 14 + 15 + 8 + 10 = 92$
 $n = 8$

$$\text{Mean}(\bar{x}) = \frac{\Sigma x}{n} = \frac{92}{8} = 11.5$$

Hence, the mean is 11.5

(ii) To find median, first we have to order given set of entries in the ascending order.

8, 10, 10, 10, 12, 13, 14, 15

$$\text{Median} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ item}$$

$$= \left(\frac{8+1}{2}\right)^{\text{th}} \text{ item}$$

$$= 4.5^{\text{th}} \text{ item}$$

$$= (4^{\text{th}} \text{ item} + 5^{\text{th}} \text{ item}) / 2$$

$$= (10 + 12) / 2$$

$$= 11$$

So, the median value is 11

2072(II) What are the desirable properties of a good average
For the data set representing the number of children
in a random sample of 10 families in neighbourhood
is : 1, 1, 2, 2, 3, 2, 5, 1, 2 and 4. Find mean, median
mode.

Solⁿ.

The desirable properties of good average are

- ① It should be rigidly defined
- ② It should be easy to calculate and simple to understand.
- ③ It should be based upon all observations.
- ④ It should be fluctuated less from sample to sample drawn from same population.

Given Set of values are
1, 1, 2, 2, 3, 2, 5, 1, 2, 4.

① Calculation of Mean.

$$\sum x = 1+1+2+2+3+2+5+1+2+4 = 23$$

$$n = 10$$

$$(\bar{x}) = \frac{\sum x}{n} = \frac{23}{10} = 2.3$$

$$\therefore \text{Mean } (\bar{x}) = 2.3$$

② Calculation of Median.

Let's first arrange the values in ascending order

1, 1, 1, 2, 2, 2, 2, 3, 4, 5

$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ item} = \left(\frac{10+1}{2} \right)^{\text{th}} \text{ item} = 5.5^{\text{th}} \text{ item}$$

2075 Five students in the Psychology class were given IQ tests. Their scores are recorded as follows: 12, 15, 10, 15 and 18. Compute Arithmetic, Geometric and Harmonic means. Also, show that $(G.M.)^2 = A.M \times H.M$

Here, given scores are
12, 15, 10, 15, 18.

(i) Calculation of Arithmetic Mean.

$$A.M = \frac{\sum x}{n} = \frac{12+15+10+15+18}{5} = 14$$

(ii) Calculation of Geometric Mean

x_i	$\log x_i$
12	1.079
15	1.176
10	1
15	1.176
18	1.255
	$\sum \log x_i = 5.686$

$$G.M = \text{Antilog of } \left[\frac{1}{n} \sum_{i=1}^n \log x_i \right]$$

$$= \text{Antilog } \left[\frac{5.686}{5} \right]$$

$$= 13.715$$

$$\therefore G.M = 13.715$$

(iii) Calculation of Harmonic Mean

$$H.M = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}} = \frac{5}{\frac{1}{12} + \frac{1}{15} + \frac{1}{10} + \frac{1}{15} + \frac{1}{18}} = 13.432$$

$$\therefore H.M = 13.432$$

2077. What do you understand by dispersion? Consider the following data, which are sample of amino acid concentration (mg/100 ml) in arthropod hemolymph.

24.6, 24.8, 23.2, 25.7, 25.1, 23.9.

Determine Range, standard deviation and coefficient of variation.

Solⁿ.

Dispersion refers to the extent to which the given set of values deviate from the central tendency such as mean, median or mode.

Given amino-acid concentration is

24.6, 24.8, 23.2, 25.7, 25.1, 23.9

(a) Range = Largest Value - Smallest Value
 $= 25.7 - 23.2$
 $= 2.5$

(b) To find standard deviation, let's calculate mean first.

$$\text{Mean } (\bar{X}) = \frac{\sum X}{n} = \frac{24.6 + 24.8 + 23.2 + 25.7 + 25.1 + 23.9}{6}$$

$$= 24.55$$

X	$\bar{X}$	$X_i - \bar{X}$	$(X_i - \bar{X})^2$
24.6		0.05	0.0025
24.8		0.25	0.0625
23.2	24.55	-1.35	1.8225
25.7		1.15	1.3225
25.1		0.55	0.3025
23.9		-0.65	0.4225

$$\sum (X_i - \bar{X})^2 = 3.9375$$

Given set of values is

4, 7, 4, 0, 5, 2, 4, 6, 4, 4

$$\begin{aligned} \text{(i) Mean} &= \frac{\sum X_i}{n} = \frac{4+7+4+0+5+2+4+6+4+4}{10} \\ &= \frac{40}{10} \\ &= 4 \end{aligned}$$

(ii) To find Median, let's first arrange given sample in increasing order.

0, 2, 4, 4, 4, 4, 4, 5, 6, 7

$$\begin{aligned} \text{Median} &= \left(\frac{n+1}{2}\right)^{\text{th}} \text{ item} \\ &= \left(\frac{10+1}{2}\right)^{\text{th}} \text{ item} \\ &= 5.5^{\text{th}} \text{ item} \\ &= (5^{\text{th}} \text{ item} + 6^{\text{th}} \text{ item}) / 2 \\ &= (4+4) / 2 \\ &= 4 \end{aligned}$$

$\therefore$ Median = 4

Range

(iii) Mode:

$$\begin{aligned} \text{Range Mode} &= \text{Largest value} - \text{Smallest value} \\ &= 5 - 0 \\ &= 5 \end{aligned}$$

(iv) Mode = 4 (since 4 is repeated most)

2079 What is meant by "measure of central tendency"? Consider the following frequency tabulation of leaf (in grams).

X_i	1.85-1.95	1.95-2.05	2.05-2.15	2.15-2.25	2.25-2.35
f_i	5	4	10	7	4

Calculate mean and standard deviation of leaf w.t.
Solution.

A measure of central tendency is a statistical concept that represents a single value within the range of data in a simple and concise manner and concentrates in the middle of the distribution.

X_i	f_i	m	fm	fm^2
1.85-1.95	5	1.9	9.5	18.05
1.95-2.05	4	2	8	16
2.05-2.15	10	2.1	21	44.1
2.15-2.25	7	2.2	15.4	33.88
2.25-2.35	4	2.3	9.2	21.16
	$N=30$		$\Sigma fm=63.1$	$\Sigma fm^2=133.19$

$$\text{Now, Mean } (\bar{X}) = \frac{\Sigma fm}{N} = \frac{63.1}{30} = 2.10$$

$$\begin{aligned} \text{Standard deviation } (\sigma) &= \sqrt{\frac{\Sigma fm^2}{N} - \left(\frac{\Sigma fm}{N}\right)^2} \\ &= \sqrt{\frac{133.19}{30} - (2.10)^2} \\ &= 0.172 \end{aligned}$$

$\therefore$ Standard deviation = 0.172
of given sample is

X	f	m	cf	fm	fm ²
165.					
176.5 - 186.5	15	181.5	15	2722.5	494133.75
186.5 - 191.5	10	189	25	1890	357210
191.5 - 196.5	10	194	35	1940	376360
196.5 - 201.5	12	199	47	2388	475212
201.5 - 206.5	6	204	53	1224	249696
206.5 - 216.5	7	211.5	60	1480.5	313125.75
	N=60			$\Sigma fm = 11645$	$\Sigma fm^2 = 2265737.5$

$$\text{Mean } (\bar{X}) = \frac{\Sigma fm}{N} = \frac{11645}{60} = 194.08$$

$$\text{Position of median} = \frac{N}{2} = \frac{60}{2} = 30$$

In c.f. just greater than 30 is 35. so,

$$L = 191.5, \quad cf = 25, \quad f = 10$$

$$\begin{aligned} \text{Median } Q_2 &= 191.5 + \frac{30 - 25}{10} \times 10 \\ &= 196.5 \end{aligned}$$

$$\begin{aligned} \sigma (\text{Standard deviation}) &= \sqrt{\frac{\Sigma fm^2}{N} - \left(\frac{\Sigma fm}{N}\right)^2} \\ &= \sqrt{\frac{2265737.5}{60} - (194.08)^2} \\ &= 9.75 \end{aligned}$$

Now, Karl Pearson's coefficient of skewness is calculated as

$$S_k(P) = \frac{3(\text{mean} - \text{median})}{\text{Standard deviation}}$$

$$= \frac{3(194.08 - 196.5)}{9.75}$$

$$= -0.744$$

Since $S_k(P) = -0.744 < 0$, the distribution is negatively skewed or left skewed

Histogram and Frequency Polygon.

The above frequency distribution is of unequal class width. So we have to calculate adjusted frequency

X	f	Adjusted frequency
176.5 - 186.5	15	7.5
186.5 - 191.5	10	10
191.5 - 196.5	10	10
196.5 - 201.5	12	12
201.5 - 206.5	6	6
206.5 - 216.5	7	3.5

Now

$$\sigma = \sqrt{\sum_{i=1}^n \frac{1}{n} (x_i - \bar{x})^2}$$

$$= \sqrt{\frac{3.935}{6}}$$

$$= 0.80$$

so, the standard deviation of given set of values is 0.80.

(c) Now, coefficient of variation is given by

$$\frac{\sigma}{(\bar{x})} \times 100 = \frac{0.8}{24.55} \times 100 = 3.29\%$$

2078 What do you mean by average in statistics? The number of sick days due to cold and flu pass year was recorded for a sample of 20 adults. The data are 4, 7, 4, 0, 5, 2, 4, 6, 4, 4. Compute mean, median, ^{range} and mode.

In statistics, an average is defined as the number that measures the central tendency of a given set of numbers.

The most common average is arithmetic mean which is defined as the ratio of sum of all observations to the number of observations.

If $a_1, a_2, a_3, \dots, a_n$ are the observations for a given sample then

$$\text{Mean } (\bar{x}) = \frac{\sum_{i=1}^n a_i}{n}, \text{ where } a_i \text{ represents individual series}$$

n = number of observations.

Now, $GM^2 = (13.715)^2 = 188.101 \dots (a)$

$AM \times HM = 14 \times 13.432 = 188.048 \sim 188.101 \dots (b)$

from (a) and (b); $GM^2 = AM \times HM$ proved.

2076. What do you mean by dispersion? Five body weights, in grams, collected from a population of rodent body weights are: 66.2, 75.1, 61.5, 71 and 69.2. Compute Mean, Variance and coefficient of Variation of these data.

→ (Defn at last)

Solution.

Given body weights in grams are;
66.2, 75.1, 61.5, 71, 69.2

(i) $Mean (\bar{X}) = \frac{\sum X}{n} = \frac{66.2 + 75.1 + 61.5 + 71 + 69.2}{5}$
 $= \frac{343}{5}$
 $= 68.6$

(ii) To find Variance, first we shall find Standard deviation

deviation			
X	$\bar{X}$	$X_i - \bar{X}$	$(X_i - \bar{X})^2$
66.2	68.6	-2.4	5.76
75.1		6.5	42.25
61.5		-7.1	50.41
71		2.4	5.76
69.2		0.6	0.36
			$\sum (X_i - \bar{X})^2 = 104.54$

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